

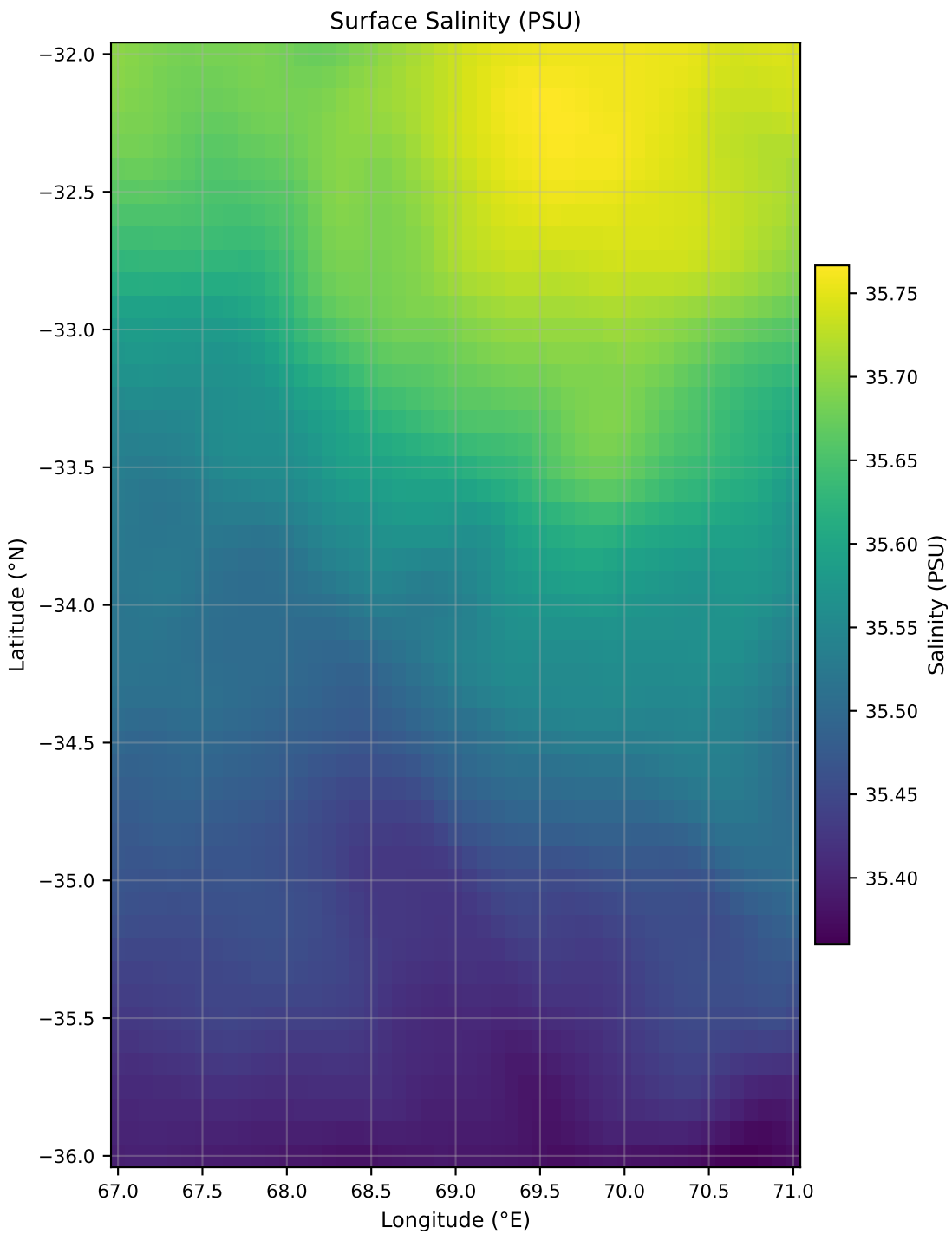
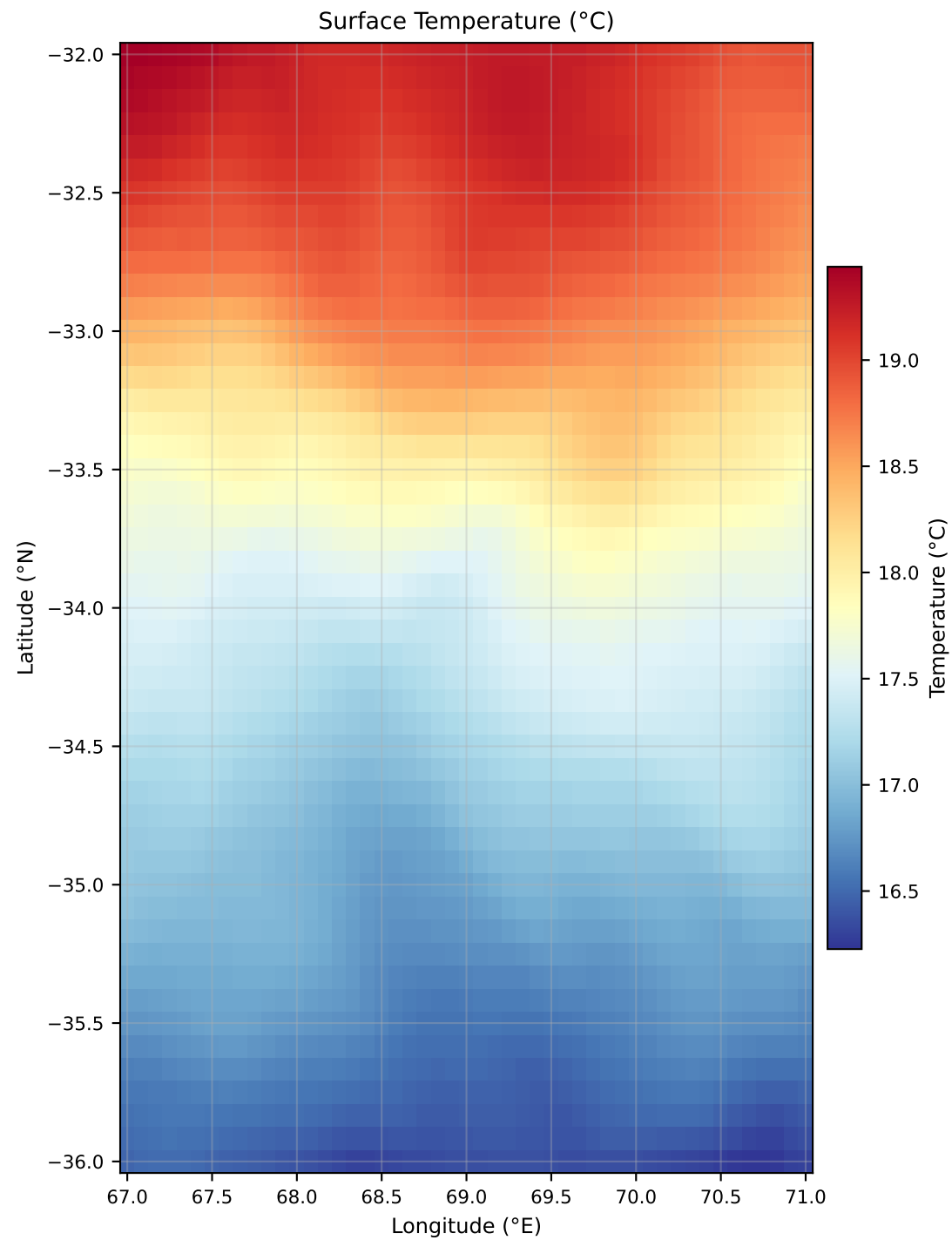
Experiment 1

INR Baseline — Dense Training · Uniform Validation

SIREN backbone · data loss only · no physics

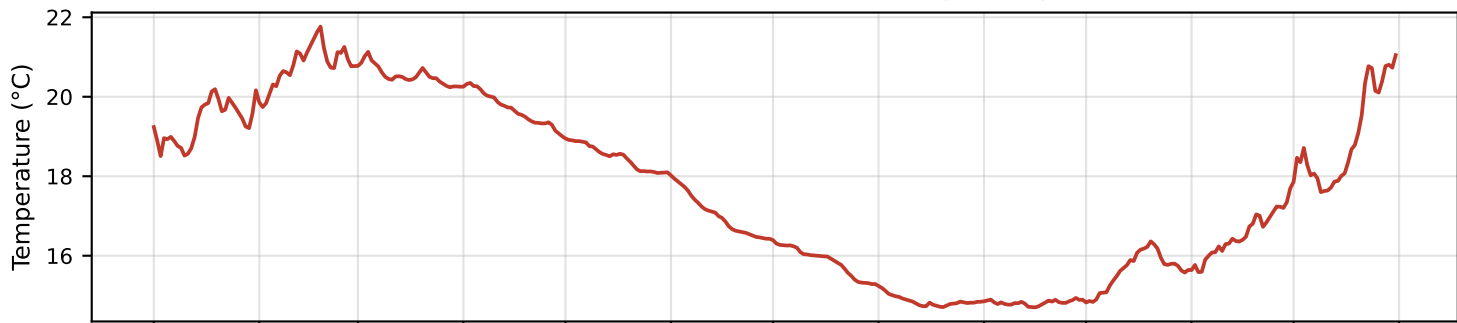
Model	SIREN 256 × 5 layers	Learning rate	1.0e-04
ω_0	20.0	Batch size	8,192
Parameters	264,962	Max epochs	2,000
		Patience	100
Split mode	uniform		
Train	70.0% (615,136 profiles)	Best epoch	1888
Validation	15.0% (131,815 profiles)	Best val loss	0.000039
Test	15.0% (131,815 profiles)		
Seed	42	Val T RMSE	0.0469 °C
		Val S RMSE	0.00609 PSU
		Test T RMSE	0.0468 °C
		Test S RMSE	0.00606 PSU

Mean Annual Sea Surface Fields

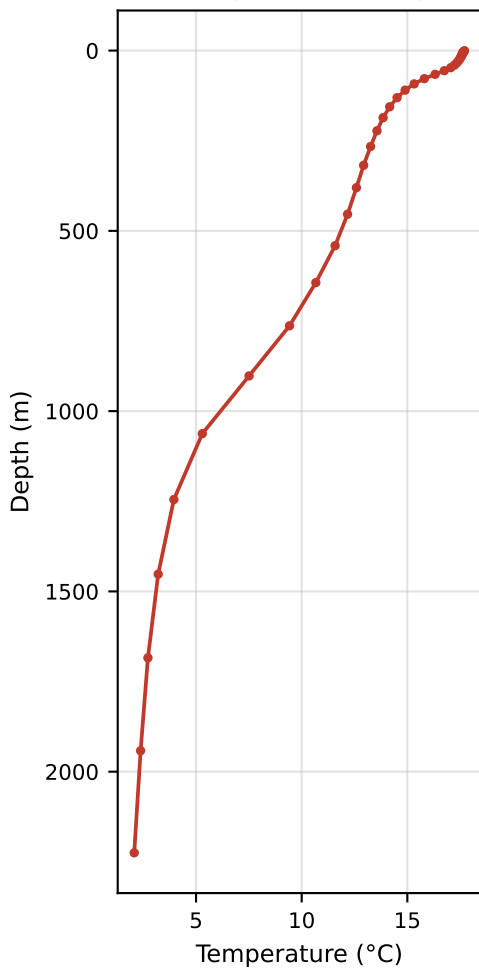


Temporal & Vertical Distribution

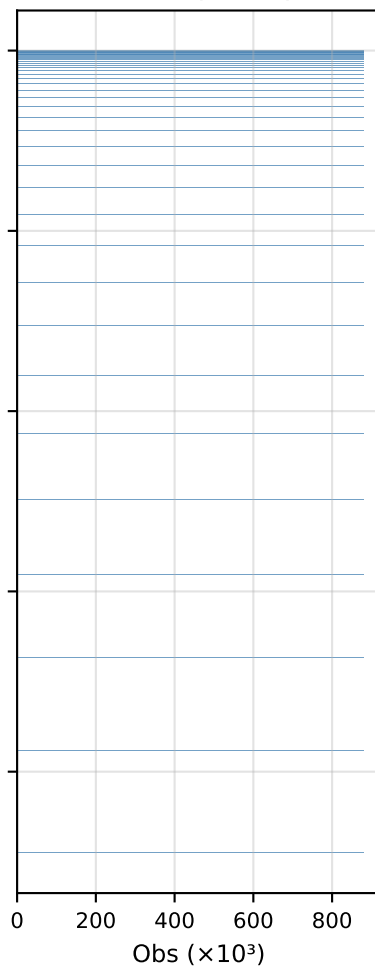
Domain-mean sea surface fields (annual)



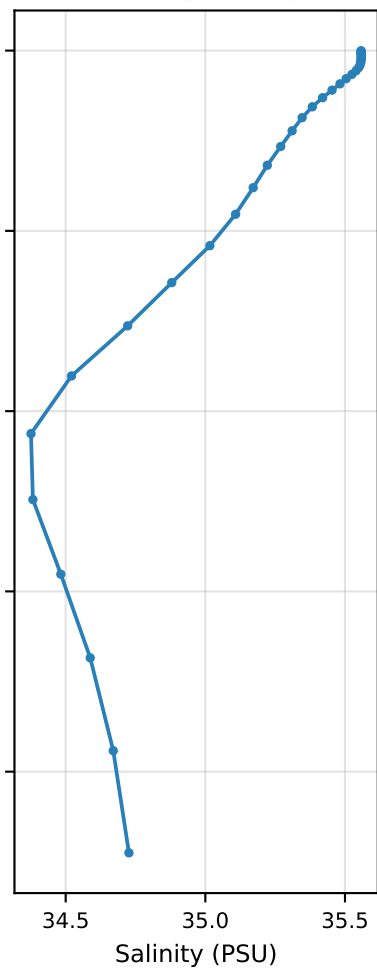
Mean Temperature (°C) profile

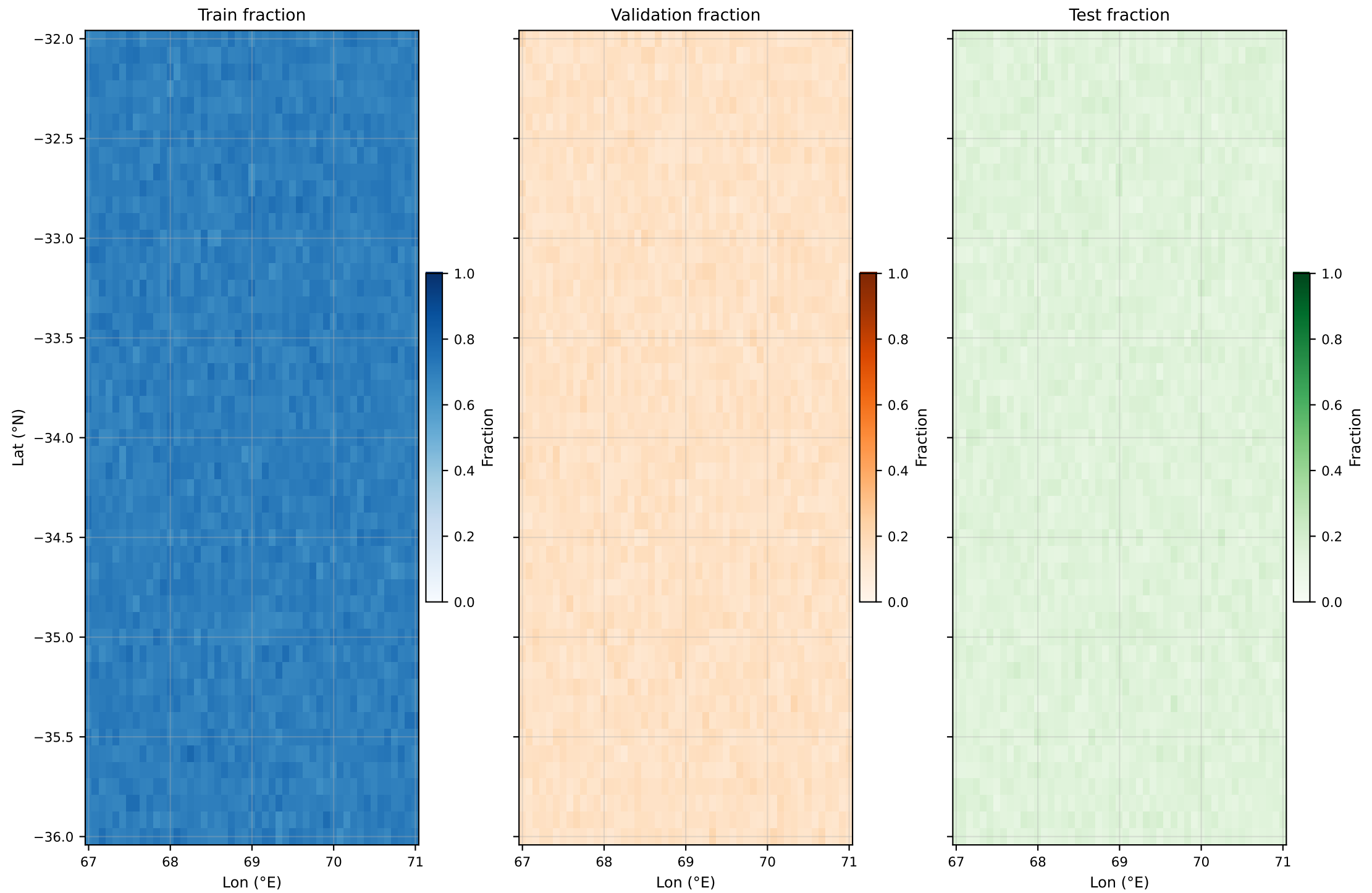


Valid obs per depth



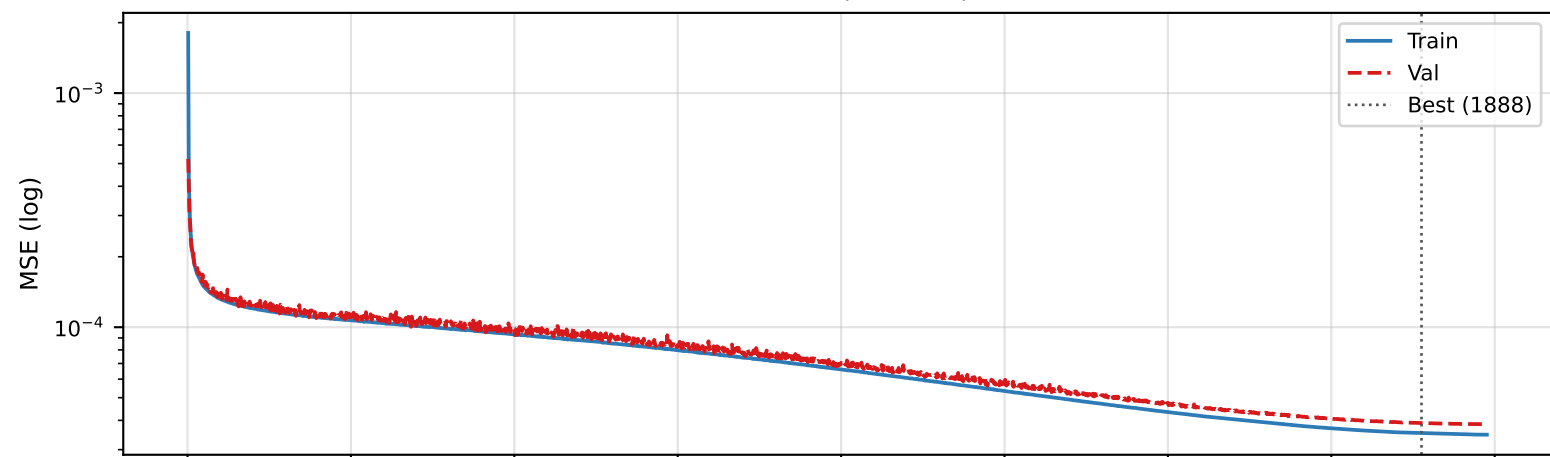
Mean Salinity (PSU) profile



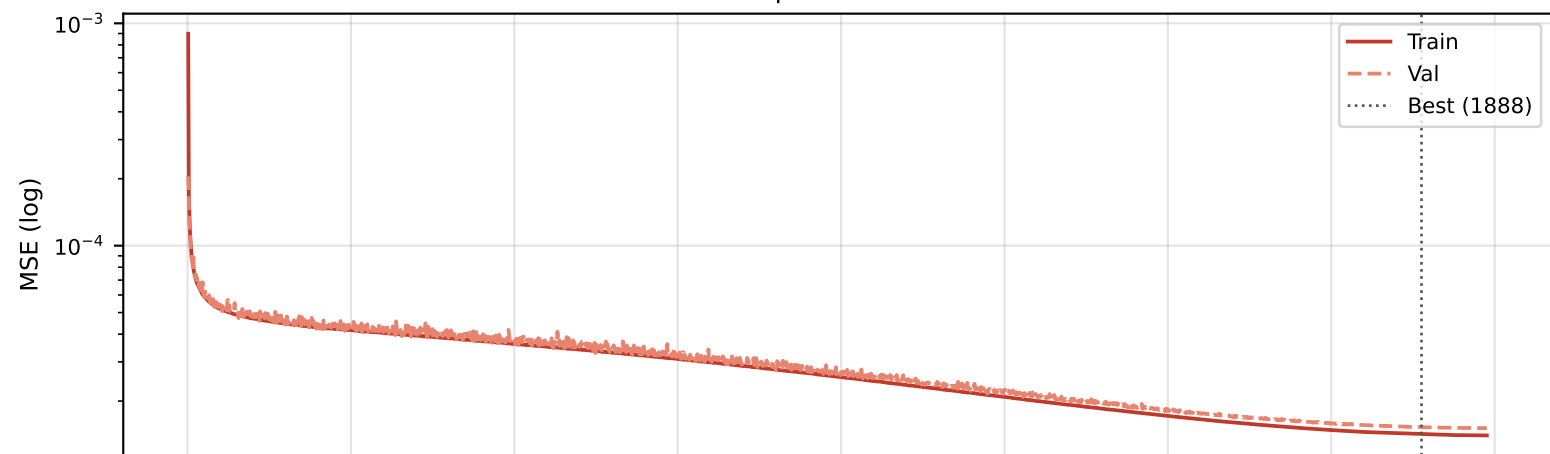


Training History

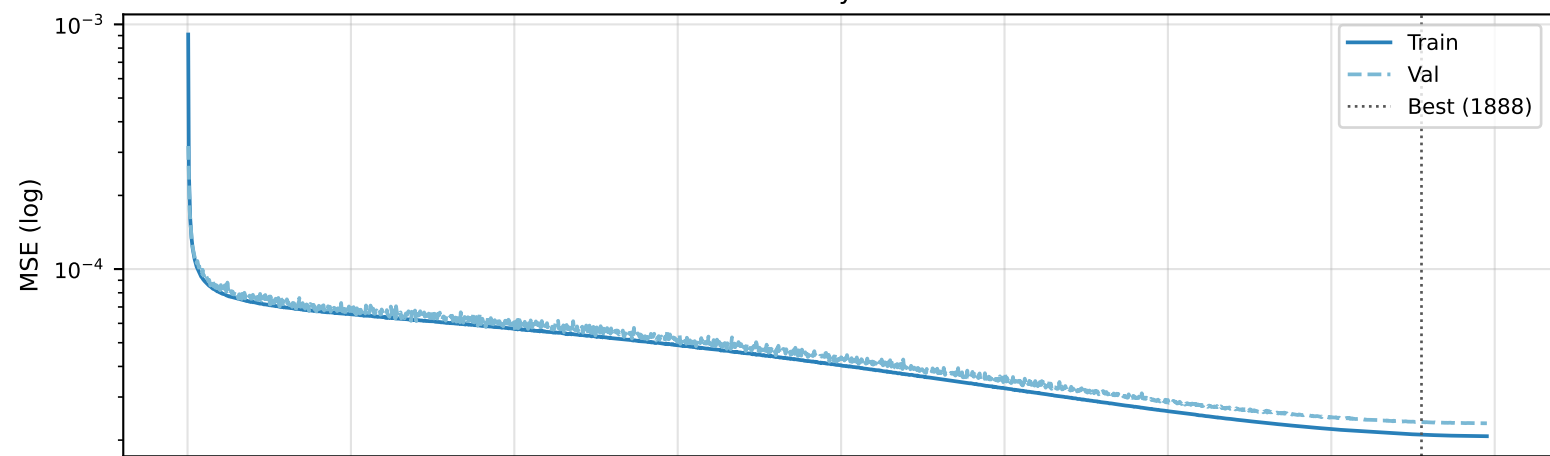
Total MSE (CT + SA)



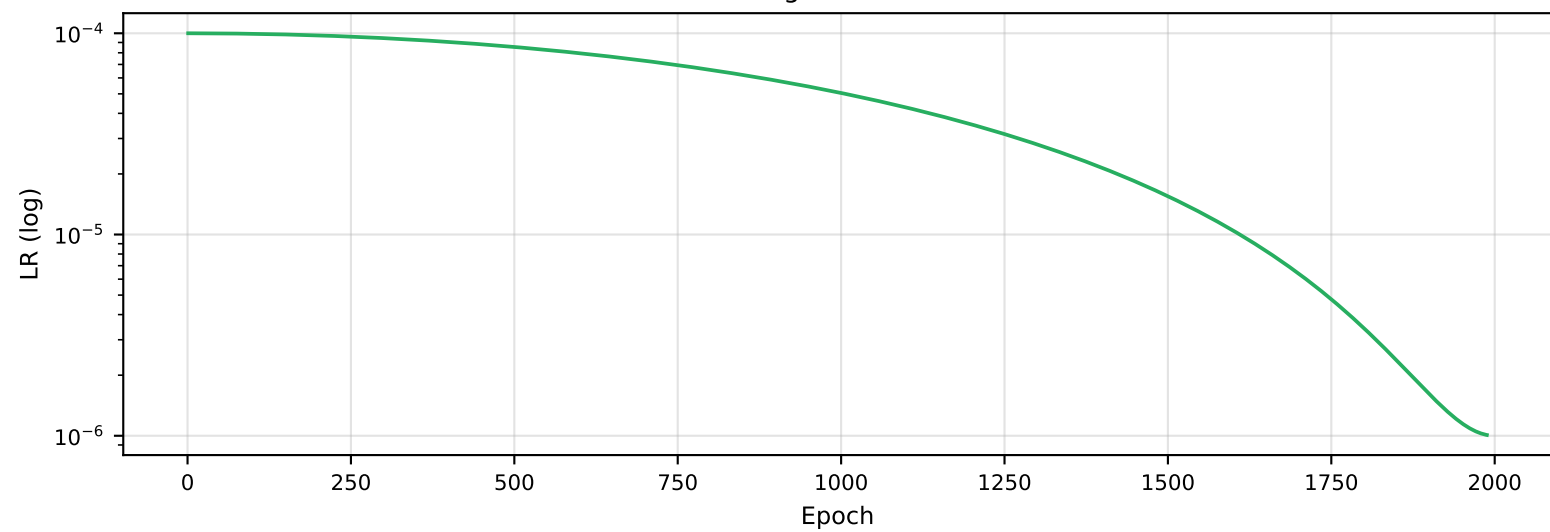
Temperature MSE



Salinity MSE

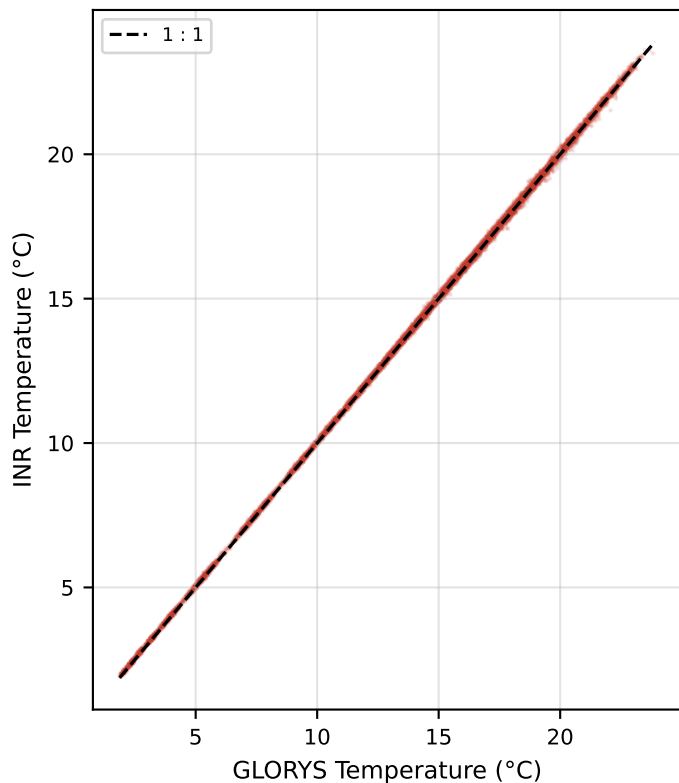


Learning rate schedule

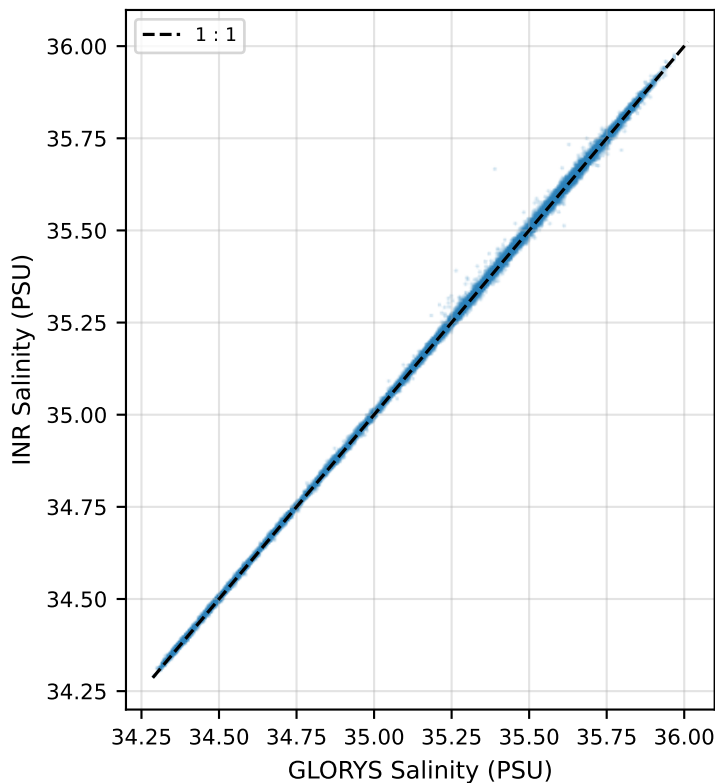


Validation — 5,404,415 observations

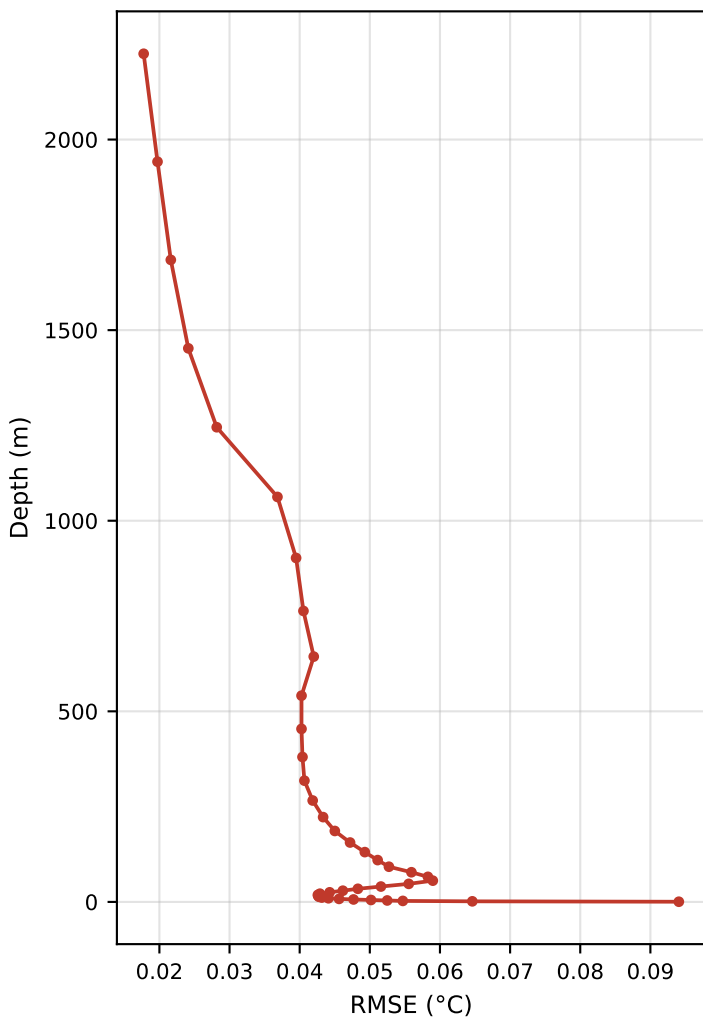
Temperature (°C)
RMSE = 0.0469 °C MAE = 0.0337 °C



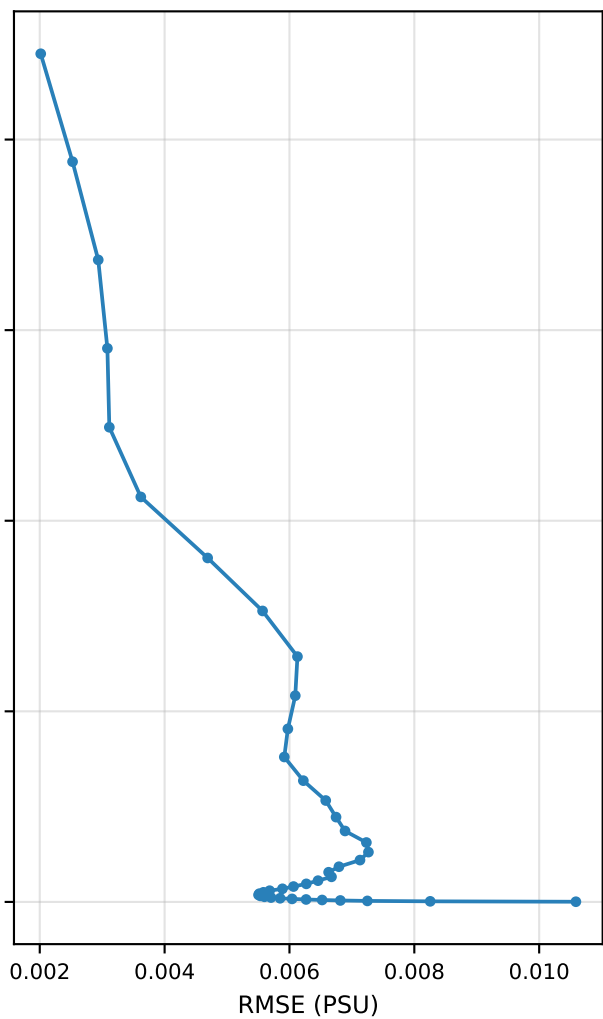
Salinity (PSU)
RMSE = 0.0061 PSU MAE = 0.0043 PSU



Temperature (°C) RMSE vs depth
Overall = 0.0451 °C

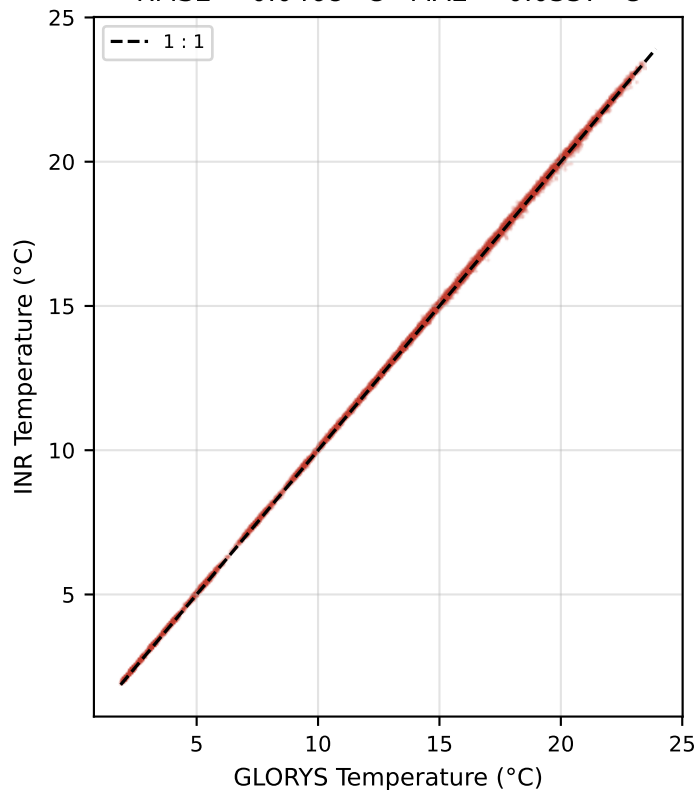


Salinity (PSU) RMSE vs depth
Overall = 0.0059 PSU

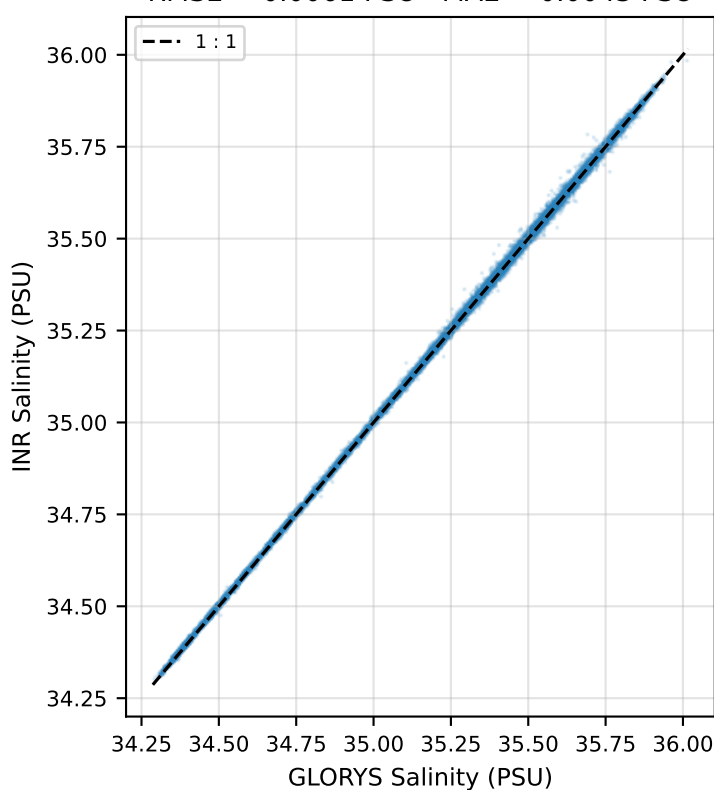


Test — 5,404,415 observations

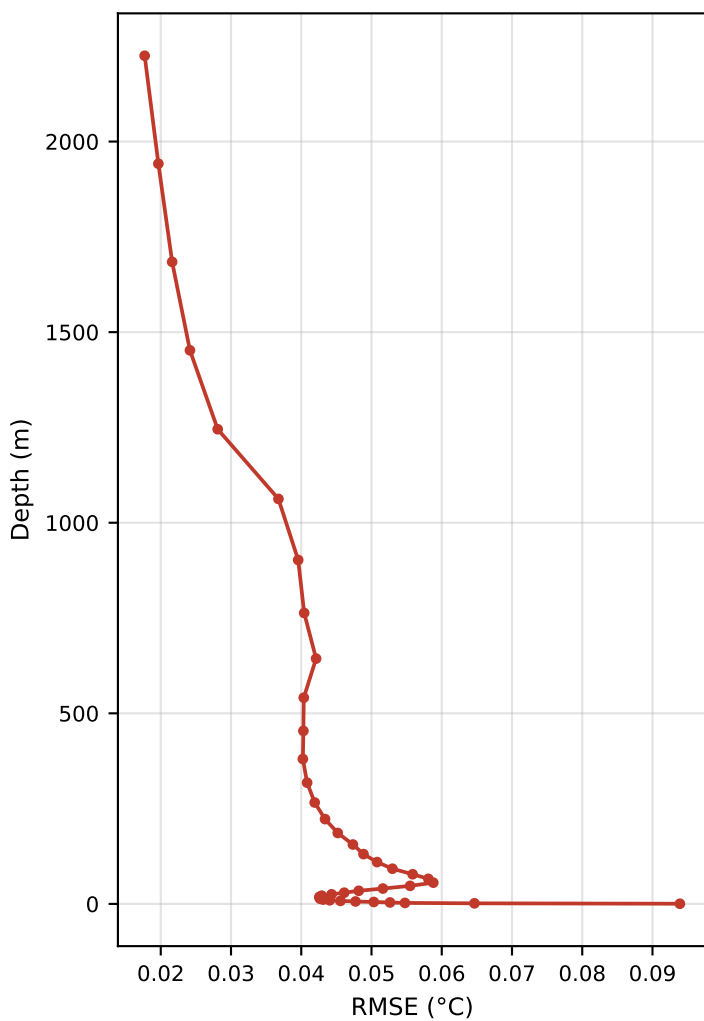
Temperature (°C)
RMSE = 0.0468 °C MAE = 0.0337 °C



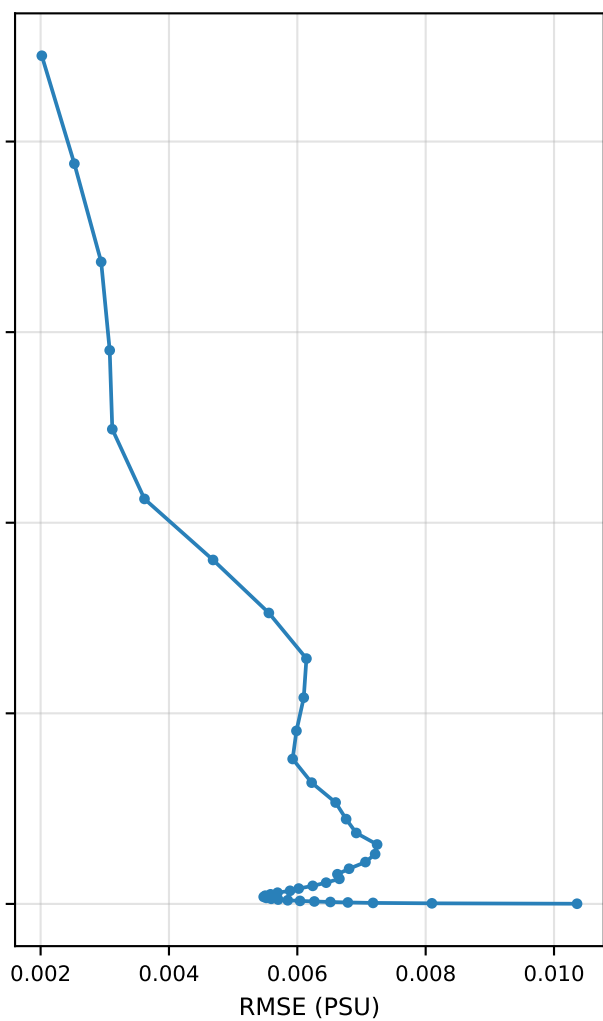
Salinity (PSU)
RMSE = 0.0061 PSU MAE = 0.0043 PSU



Temperature (°C) RMSE vs depth
Overall = 0.0451 °C



Salinity (PSU) RMSE vs depth
Overall = 0.0059 PSU



Summary — Final Metrics

Set	Temperature (°C) RMSE	Temperature (°C) MAE	Salinity (PSU) RMSE	Salinity (PSU) MAE
Validation	0.0469	0.0337	0.00609	0.00427
Test	0.0468	0.0337	0.00606	0.00426

